SUNIL A M

SOFTWARE DEVELOPER | AI / GENERATIVE AI SPECIALIZATION

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PROFESSIONAL SUMMARY

Software Developer with a strong foundation in full-stack development, backend engineering, and Artificial Intelligence/Machine Learning. Hands-on experience with Python, JavaScript, React, Node.js, Django, Flask, REST APIs, MongoDB, and MySQL. Experienced in machine learning, computer vision, data preprocessing, feature engineering, model evaluation, and AI-powered application development. Developing practical skills in Generative AI, Large Language Models (LLMs), LLM APIs, Retrieval-Augmented Generation (RAG), embeddings, vector databases, AI agents, tool calling, MCP, Docker, and cloud deployment. Seeking Software Developer opportunities where software engineering and AI skills can be combined to build scalable intelligent applications.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Frontend: HTML, CSS, React, Next.js, Tailwind CSS

Backend: Node.js, Express.js, Django, Flask, FastAPI, REST APIs

AI / Machine Learning: Machine Learning, Data Preprocessing, Feature Engineering, Model Evaluation, Computer Vision, OpenCV, MobileFaceNet, NumPy, Pandas, Scikit-learn, PyTorch, Hugging Face Transformers

Generative AI: Large Language Models (LLMs), Prompt Engineering, OpenAI API, Gemini API, Anthropic API, Embeddings, Retrieval-Augmented Generation (RAG), Vector Search, Function Calling, Tool Calling, Structured Outputs, AI Agents, Model Context Protocol (MCP)

AI Frameworks: LangChain, LlamaIndex, RAG Pipelines, Semantic Search, AI API Integration

Databases: MongoDB, MySQL, PostgreSQL, Redis, Vector Databases

Software Engineering: Object-Oriented Programming, Data Structures and Algorithms, API Integration, JWT, Authentication, Role-Based Access Control, Debugging, Testing

DevOps / Cloud: Git, GitHub, Docker, CI/CD Fundamentals, AWS Fundamentals, Environment Variables

Deployment Tools: VS Code, Postman, Firebase, Vercel, Netlify, Render, Railway

PROFESSIONAL EXPERIENCE

AI & MACHINE LEARNING INTERN — Spherenex | Feb 2026 – May 2026

- Built and trained machine learning models using Python for real-world datasets.
- Performed data preprocessing and feature engineering for machine learning applications.
- Evaluated machine learning models to assess performance and improve results.
- Developed and deployed multiple machine learning projects.
- Applied Python-based machine learning techniques to solve practical problems.

PROJECTS

AI-POWERED DOCUMENT INTELLIGENCE & AGENT PLATFORM | Python, FastAPI, LLM APIs, RAG, Embeddings, Vector Database, AI Agents, Tool Calling, MCP, PostgreSQL, Docker

- Developed an AI-powered document intelligence application for answering questions from user-provided documents using Large Language Models.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with document chunking, embeddings, vector search, and context-aware LLM responses.
- Integrated LLM APIs and structured outputs for reliable AI response generation and application workflows.
- Implemented semantic search with embeddings and a vector database to retrieve relevant document context.
- Developed an AI agent capable of selecting and using tools through function calling and tool calling.
- Integrated external tools and APIs into agent workflows and explored Model Context Protocol (MCP) for AI-to-tool connectivity.
- Built REST APIs using FastAPI and used PostgreSQL for application data and conversation management.
- Containerized the application using Docker for consistent development and deployment.

BLOCKCHAIN & AI-POWERED HEALTHCARE MANAGEMENT SYSTEM | Python, Django, MongoDB, Blockchain, REST APIs, Machine Learning

- Developed a secure healthcare management platform using blockchain for tamper-resistant storage and integrity of medical records.
- Integrated machine learning models to analyze patient data and generate predictive health insights for early disease detection.
- Developed REST APIs and implemented role-based authentication and secure data access.
- Designed a responsive web application for efficient patient record management.

AI-BASED SMART ATTENDANCE SYSTEM | Python, Flask, OpenCV, MobileFaceNet

- Developed an AI-based attendance system using real-time face recognition for automated attendance management.
- Implemented facial recognition, duplicate attendance prevention, automated attendance recording, and CSV export.
- Built a web interface for student registration and real-time attendance tracking with voice announcements.

ONLINE JOB PORTAL | JavaScript, Node.js, Express.js, MongoDB, REST APIs, JWT

- Developed a full-stack job portal for job posting, searching, applications, and application tracking.
- Implemented REST APIs, JWT-based authentication, role-based access control, and MongoDB data management.

EDUCATION

B.E. IN COMPUTER SCIENCE — Bahubali College of Engineering, Shravanabelagola, Karnataka | Aug 2022 – Jun 2026 | CGPA: 8.2

PRE-UNIVERSITY — Govt PU College, Chikkamagaluru, Karnataka | May 2020 – Jul 2022 | Percentage: 64.5%

RESEARCH PUBLICATION

Blockchain and AI-Powered Healthcare Management System — International Journal of Innovative Research in Technology (IJIRT). ISSN 2349-6002 | Vol. 12 | Issue 8 | January 2026.

CERTIFICATIONS

- Website Development Tutorial — Infosys Springboard | February 2026
- Introduction to MERN Stack — Simplilearn SkillUP | April 2026 | Certificate Code: 9924435

ADDITIONAL LEARNING

Generative AI and LLM Application Development; RAG and Vector Search; AI Agents, Function Calling, Tool Calling and MCP; Docker and Cloud Deployment Fundamentals; Modern AI Application Architecture.